

Group members



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Your answer NOT SUBMITTED

Assignment

Part 1: Regression and Classification Activity

Objective

This part builds on your previous group work and focuses on applying regression and classification models to solve a practical problem. You will explore datasets, split data for training and testing, apply models, evaluate their performance, and document your findings while considering technical and ethical aspects.

Instructions

Step 1: Revisiting Your Problem Statement

1. Review Your Group's Chosen Problem Statement:

- Revisit the problem statement you developed in the previous assignment.
- Ensure it is suitable for both regression and classification tasks.
- If necessary, temporarily consider a related problem statement to align with the objectives of this assignment.

2. Verify Dataset Suitability:

- Ensure your dataset is suitable for regression and at least two classification tasks.

- If your current dataset does not meet these requirements, identify a similar dataset that does.
 - Use reputable sources such as:
 - Kaggle
 - UCI Machine Learning Repository
 - Google Dataset Search
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Step 2: Data Splitting and Preparation

1. Split Your Dataset:

- Divide your dataset into **training**, **validation**, and **test** subsets.
- Recommended split ratios:
 - Training set: 70–80%
 - Validation set (optional): 10–15%
 - Test set: 20–30%

2. Explain the Splitting Process:

- Provide an explanation of the splitting strategy used (e.g., random split, stratified split).
- Include the rationale for your choice of splitting method.

3. Prepare Your Data:

- Perform any necessary preprocessing steps, including:
 - Handling missing values (e.g., imputation).
 - Scaling and normalizing features.
 - Encoding categorical variables (e.g., one-hot encoding).
 - Document and justify each preprocessing step.
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Step 3: Exploring Regression

1. Apply a Regression Model:

- Choose one regression model: **Linear Regression** or **Polynomial Regression**.
- Implement the model to predict a continuous target variable relevant to your problem.

2. Evaluate the Model:

Visualize the results using scatter plots, fitted lines, or other relevant charts.

3. Document Your Findings:

- Provide a brief analysis of the model's performance.
 - Highlight any challenges or observations during the implementation.
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Step 4: Exploring Classification

1. Apply At Least Two Classification Models:

- Select at least two classification models from the following:
 - Logistic Regression
 - Decision Trees
 - Support Vector Machines (SVM)
 - k-Nearest Neighbors (k-NN)
 - Naive Bayes
- Use the models to classify your dataset based on a discrete target variable.

2. Visualize the Results:

- Provide visualizations of decision boundaries, confusion matrices, or other relevant plots.
- Include any preprocessing steps specific to classification (e.g., feature scaling for SVM).

3. Document Insights:

- Summarize the strengths and weaknesses of each model based on your results.
- Discuss the most effective model for your problem and why.

Step 5: Ethical and Practical Considerations

1. Ethical Risk Identification:

- Identify any new ethical risks associated with your AI solution (e.g., bias, privacy, transparency).
- Use the guidelines from the previous module to assess the severity and likelihood of the risk.

2. Propose Mitigation Strategies:

- Outline steps to mitigate the new identified risks (if any).

3. Evaluate Model Feasibility:

- Reflect on the practicality of your solution, including:
 - Dataset quality and availability.
 - Resource requirements for implementation.

Submission Requirements

Group Report

1. Content:

Combine all findings into a single document, including:

- Problem statement (refined, if applicable).
- Dataset description and analysis.
- Explanation of the data splitting process and rationale.
- Regression and classification results with visualizations.
- New ethical risk assessment and mitigation strategies (if any).
- Key insights and conclusions.

2. Structure:

- Use clear headings and subheadings to structure your report.

Code and Notebooks

1. Content:

- Submit Python code and/or Jupyter notebooks used for your analyses.
- Ensure the code is well-commented and easy to understand.

2. Visualizations:

- Include at least one visualization for each model used.
- Clearly label and annotate all charts and graphs.

3. Screenshots:

- Provide screenshots of outputs to demonstrate the results of your tasks.

Resources and Support

- Refer to the slides in **Fundamental Algorithms** for detailed guidance.
- Explore datasets using the recommended sources:
 - Kaggle
 - UCI Machine Learning Repository
 - Google Dataset Search
- Use Python libraries such as:
 - scikit-learn
 - pandas
 - matplotlib
 - seaborn

Part 2: Unsupervised Learning Activity

Objective

This part of the assignment focuses on applying unsupervised learning techniques—clustering and dimensionality reduction—to analyze unlabeled datasets. You will explore these methods, evaluate their effectiveness, and visualize results to derive insights.

Instructions

Step 1: Clustering Analysis

1. Choose Two Clustering Models:

- Select two models from the following:
 - K-Means
 - DBSCAN
 - Hierarchical Clustering

2. Apply Each Model to Your Dataset:

- Analyze the dataset using the chosen models.
- Tune hyperparameters (e.g., `n_clusters` for K-Means, `eps` for DBSCAN).

3. Evaluate Clustering Performance:

- Calculate clustering quality using metrics like:
 - Silhouette Score.
 - Davies-Bouldin Index.
- Include a comparative analysis of the models' strengths and weaknesses for your dataset.

4. Visualize Results:

- Use 2D scatter plots, PCA-reduced visualizations, or dendrograms to represent clusters.
- Clearly label clusters and centroids (if applicable).

Step 2: Dimensionality Reduction Analysis

1. Apply PCA:

- Reduce the dataset's dimensions using PCA.
- Choose the optimal number of components to explain at least 85% of the variance.
- Visualize the PCA-reduced data in 2D.

2. Apply t-SNE (Optional):

- Perform t-SNE to explore non-linear patterns in the dataset.
- Compare the visual representation with PCA.

3. Document Key Findings:

- Summarize the benefits and limitations of using PCA and t-SNE for dimensionality reduction on your dataset.

Step 3: Comparative Analysis and Insights

1. Compare Models and Techniques:

- Discuss how clustering (e.g., K-Means vs. DBSCAN) complements dimensionality reduction (e.g., PCA).
- Highlight key insights obtained from combining these techniques.

2. Practical Challenges:

- Document any difficulties in hyperparameter tuning, convergence issues, or visualization.

3. Real-World Applications:

- Provide examples of how similar clustering and dimensionality reduction techniques are applied in real-world scenarios (e.g., customer segmentation, fraud detection).

Step 4: Ethical and Practical Considerations

1. Assess Ethical Risks:

- Discuss potential ethical concerns (e.g., bias in clustering results, misinterpretation of dimensionality reduction).
- Evaluate risks using guidelines provided in the module.

2. Propose Mitigation Strategies:

- Suggest methods to address identified risks, such as anonymization, proper data preprocessing, and transparent reporting.

3. Reflect on Feasibility:

- Analyze the practicality of the solutions, considering dataset quality, resource requirements, and interpretability.

Submission Requirements

1. Group Report:

- Include the following:
 - Problem statement and dataset description.
 - Clustering analysis results (metrics, visualizations).
 - Dimensionality reduction results (PCA and optional t-SNE).
 - Comparative analysis and practical insights.
 - Ethical and practical considerations.

2. Code and Notebooks:

- Submit well-documented Python code or Jupyter notebooks.
- Include comments explaining critical steps (e.g., hyperparameter tuning, evaluation metrics).
- Attach screenshots of outputs where relevant.

3. Visualizations:

- Provide clear and labeled plots for:
 - Clustering results.
 - PCA or t-SNE visualizations.
 - Use distinct color coding and annotations.
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Part 3: Reinforcement Learning Activity – Q-Learning Exploration

Objective

This part of the assignment focuses on exploring the fundamentals of Reinforcement Learning (RL) through hands-on activities with a provided Q-learning demonstration programs. The goal is to understand how RL agents learn, experiment with key parameters, and reflect on the challenges and applications of RL in real-world scenarios.

Instructions

Step 1: Running the Q-Learning Program

1. Set up and execute the provided Python programs:

- Use the Q-learning program shared in the course materials to simulate agent learning in a grid-world environment.

- Ensure the program runs correctly, and familiarize yourself with its interface and controls.

2. **Modify the environment:**

- Change the grid size, start and goal positions, and add/remove obstacles.
- Observe how these changes influence the agent's ability to learn optimal paths.

3. **Experiment with hyperparameters:**

- Adjust the following parameters:
 - **Learning rate (α):** Controls how quickly the agent updates its knowledge based on new experiences.
 - **Discount factor (γ):** Determines how much the agent prioritizes long-term rewards over immediate rewards.
 - **Exploration rate (ϵ):** Balances exploration (trying new actions) and exploitation (using known optimal actions).
- Run multiple training sessions with different parameter configurations and observe how they affect the learning process.

4. **Use visualization tools:**

- Utilize the program's features, such as "View Map" and "View Q-Table," to monitor the agent's progress and identify patterns in its learning behavior.

Step 2: Observing the Q-Learning Behavior

1. **Track the agent's real-time movement:**

- Analyze how the robot navigates the grid and adapts to reach the goal.
- Observe whether the agent's path becomes more efficient over time.

2. Monitor Q-table updates:

- Watch how Q-values evolve over multiple episodes.
- Identify which states and actions the agent prioritizes as it learns.

3. Document your observations:

- Record significant moments in the agent's learning process, such as:
 - Initial attempts vs. later optimal paths.
 - The effect of obstacles or grid layout on the agent's learning.
 - The stabilization or divergence of Q-values over time.

Step 3: Reflection and Analysis

1. Answer the following questions in your report:

- How does the agent adapt its behavior to reach the goal?
- What impact do hyperparameters (α , γ , ϵ) have on the learning efficiency and path selection?
- How do grid layout, obstacles, and goal placement influence learning outcomes?

2. Analyze challenges and insights:

- Discuss the potential real-world challenges of applying RL, such as scalability to large environments or handling continuous state/action spaces.
- Reflect on how reward structures and parameter tuning affect learning success.

Step 4: Ethical and Practical Considerations

1. Identify potential risks:

- Consider ethical issues, such as:
 - Reward hacking: Unintended behaviors caused by poorly designed rewards.
 - Safety concerns: Risks of the agent making unsafe or suboptimal decisions in real-world scenarios.
- Discuss the scalability challenges of RL in large or dynamic environments.

2. Propose mitigation strategies:

- Suggest ways to design safe and robust RL systems, such as:
 - Careful reward function design.
 - Regular monitoring and retraining to adapt to changing environments.

Submission Requirements

1. Group Report:

- Include:
 - A description of your modifications to the environment and hyperparameters.
 - Observations of agent behavior, Q-table updates, and learning efficiency.
 - Reflections on challenges, ethical risks, and real-world implications of RL.
- Use clear headings and subheadings to organize your report.

2. Code and Outputs:

- Submit the Python script or Jupyter notebook used for this activity.
- Include comments explaining changes made to the environment or hyperparameters.

- Attach screenshots or output logs showing key moments in the agent's learning process.

3. **Reflections:**

- Provide a brief section summarizing your group's key takeaways and insights from the activity.